

class17

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Table of contents

Setup	1
PCA	2
Plot of Transcriptomic Profiles	2

Setup

```
library(tximport)

# setup the folder and filenames to read
folders <- dir(pattern="SRR21568*")
samples <- sub("_quant", "", folders)
files <- file.path( folders, "abundance.h5" )
names(files) <- samples

txi.kallisto <- tximport(files, type = "kallisto", txOut = TRUE)
```

1 2 3 4

```
head(txi.kallisto$counts)
```

	SRR2156848	SRR2156849	SRR2156850	SRR2156851
ENST00000539570	0	0	0.00000	0
ENST00000576455	0	0	2.62037	0
ENST00000510508	0	0	0.00000	0
ENST00000474471	0	1	1.00000	0
ENST00000381700	0	0	0.00000	0
ENST00000445946	0	0	0.00000	0

Number of transcripts for each sample:

```
colSums(txi.kallisto$counts)
```

```
SRR2156848 SRR2156849 SRR2156850 SRR2156851
      2563611      2600800      2372309      2111474
```

How many transcripts detected in at least one sample:

```
sum(rowSums(txi.kallisto$counts)>0)
```

```
[1] 94561
```

Filter out transcripts with no reads:

```
to.keep <- rowSums(txi.kallisto$counts) > 0
kset.nonzero <- txi.kallisto$counts[to.keep,]
```

Filter out transcripts with no change:

```
keep2 <- apply(kset.nonzero,1,sd)>0
x <- kset.nonzero[keep2,]
```

PCA

Run PCA

```
pca <- prcomp(t(x), scale=TRUE)
summary(pca)
```

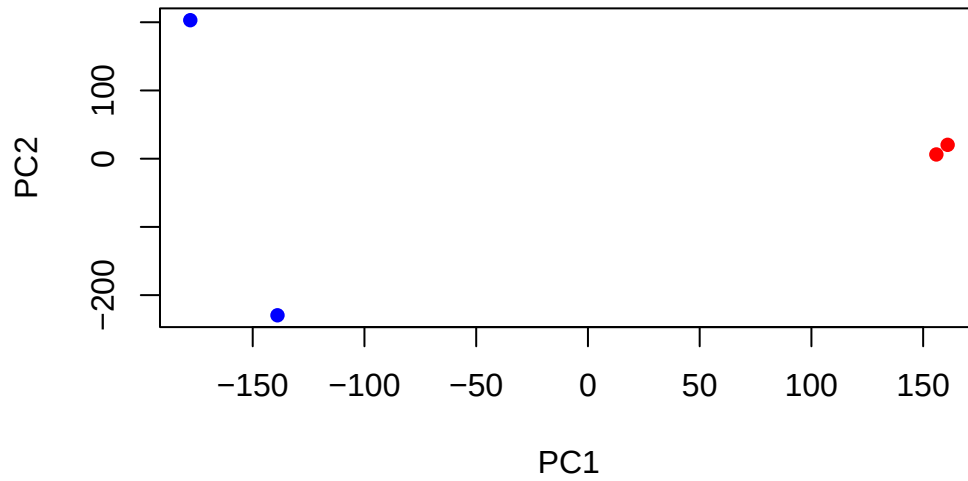
Importance of components:

	PC1	PC2	PC3	PC4
Standard deviation	183.6379	177.3605	171.3020	1e+00
Proportion of Variance	0.3568	0.3328	0.3104	1e-05
Cumulative Proportion	0.3568	0.6895	1.0000	1e+00

Plot of Transcriptomic Profiles

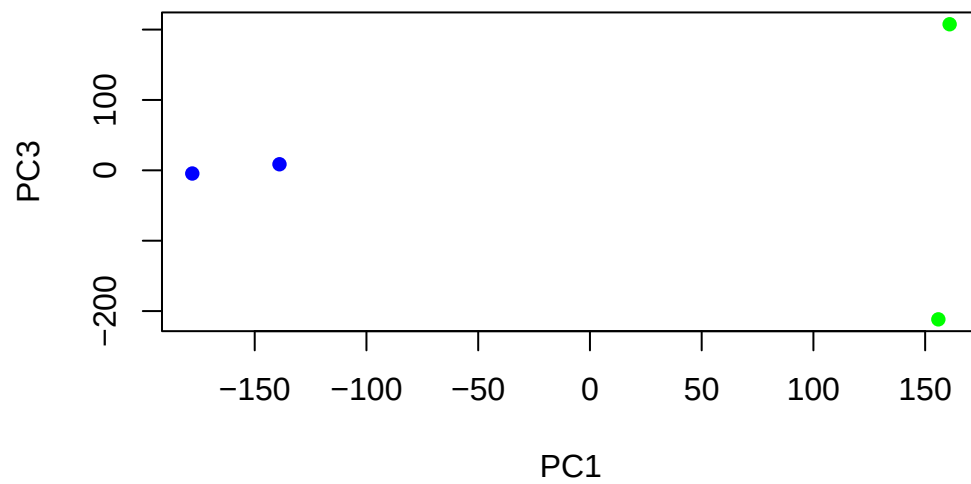
PC1 vs PC2:

```
plot(pca$x[,1], pca$x[,2],
     col=c("blue", "blue", "red", "red"),
     xlab="PC1", ylab="PC2", pch=16)
```



PC1 vs PC3:

```
plot(pca$x[,1], pca$x[,3],
     col=c("blue", "blue", "green", "green"),
     xlab="PC1", ylab="PC3", pch=16)
```



PC2 vs PC3:

```
plot(pca$x[,2], pca$x[,3],  
     col=c("red","red","green","green"),  
     xlab="PC2", ylab="PC3", pch=16)
```

